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TRACE-RAF: Trust-Gated Residual Analog Correction with Evidence-Preserving Retrieval for Auditable PM2.5 Forecasting

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ABSTRACT Short-horizon PM2.5 forecasting must combine pollutant persistence, meteorology, calendar effects, sparse environmental events, and temporal distribution shift without introducing leakage. This study proposes TRACE-RAF, a supervised Trust-gated Residual Analog Correction method for auditable 24-hour forecasting. A Source-Audited Context Builder aligns official EPA PM2.5, NOAA GHCNh weather, calendar, and hash-verified NOAA Storm Events records. A Leakage-Safe Event-Context Retriever enforces a candidate-target embargo, while TRACE-RAF retrieves out-of-fold base-model residual trajectories rather than raw futures. An 18-variable reliability gate scales each correction using retrieval quality, model disagreement, correction spread, event context, and distribution-shift evidence. A Drift-Aware Forecast and Evidence Head preserves prediction-level evidence and diagnostics. Evaluation covers Los Angeles County from 2019 through 2025 using a 168-hour lookback and chronological splits of 45,784 training, 7,950 validation, and 2,460 test origins. TRACE-RAF obtains MSE 40.062, MAE 4.134, and RMSE 6.329. It improves MSE by 4.63% over context XGBoost and is within 0.07% of context LightGBM, although Ridge remains best overall at MSE 39.144. TRACE-RAF reduces MAE by 0.021 relative to its base ensemble, but this nominal effect does not survive Holm correction. Validation assigns zero weight to raw-analogue fusion and a global residual correction, whereas ungated residual transfer records MSE 44.959. The event-free TRACE-RAF variant is nearly identical, so an event-specific gain is not established. Event-conditioned output-space fusion significantly worsens the tested frozen Chronos-Bolt MAE after multiplicity adjustment; this finding does not generalize to retrieval-augmented time-series foundation models. Because Storm Events lacks machine-readable publication timestamps, all event-conditioned results remain retrospective sensitivity analyses.

INDEX TERMS Air quality forecasting, PM2.5, retrieval-augmented forecasting, residual correction, time-series foundation models, distribution-shift diagnostics, auditable forecasting.

I. INTRODUCTION

FINE particulate matter (PM2.5) is a major urban air-quality risk because it is associated with respiratory and cardiovascular health effects [1]. Accurate short-term PM2.5 forecasts can support public-health alerts, exposure reduction, transport planning, and environmental operations. In Los Angeles County, however, PM2.5 dynamics are difficult to model with pollutant history alone. Concentrations can change with wind speed, boundary-layer behavior, rainfall, seasonal activity, holidays, wildfire smoke, and unusual storm or high-wind events. These factors make the forecasting problem both temporal and contextual.

Recent time-series foundation models (TSFMs) aim to improve forecasting generalization by pre-training on large time-series corpora and transferring the learned forecasting ability to new domains. Representative models include TimesFM, Moirai, MOMENT, Chronos, and Tiny Time Mixers [2]–[6]. In parallel, retrieval-augmented generation has shown that non-parametric memory can help foundation models access relevant external knowledge at inference time [7], [8]. TimeRAF combines these two ideas for zero-shot time-series forecasting by retrieving relevant historical sequences from a curated knowledge base and integrating them into a frozen TSFM through Channel Prompting [9].

Air-quality forecasting raises a domain-specific retrieval question that is not fully addressed by numerical time-series retrieval alone. A historical PM2.5 window may look similar to the current window but occur under different weather or event conditions. Conversely, a window with moderate numerical similarity may be more informative if it shares high-wind, rainfall, holiday, smoke, or stagnation context. This motivates a retrieval framework that treats source records, weather, calendar information, and distribution-shift indicators as part of the forecasting evidence rather than as unrelated metadata.

This study introduces TRACE-RAF, a Trust-gated Residual Analog Correction mechanism for auditable PM2.5 forecasting with event context evaluated as an auxiliary evidence channel under temporal distribution shift. The framework uses retrieval augmentation in an environmental forecasting setting while remaining architecturally distinct from TimeRAF's learned retriever and Channel Prompting mechanism. Four modules define the implemented pipeline: a Source-Audited Context Builder (SACB), a Leakage-Safe Event-Context Retriever (LSER), the TRACE residual-correction core, and a Drift-Aware Forecast and Evidence Head (DFEH). Together they address four questions: how official records can be converted into retrieval-compatible context; how candidates can be selected without temporal leakage; how analogue residuals can correct a strong supervised forecast without replacing it; and how each forecast can retain auditable retrieval, event, drift, and uncertainty evidence.

The main contributions are as follows:

- *Source-Audited Context Builder:* a reproducible mechanism that aligns EPA PM2.5, NOAA weather, calendar, and hash-verified NOAA Storm Events records while preserving source identity, coverage checks, and the declared event-availability assumption.
- *Leakage-Safe Event-Context Retriever:* a historical analogue mechanism that combines PM2.5 shape, weather, calendar, and event similarity after enforcing an embargo under which every candidate target ends before the query lookback begins.
- *TRACE residual correction:* an out-of-fold residual memory and validation-trained reliability gate that applies analogue corrections selectively to a convex LightGBM–XGBoost context ensemble.
- *Drift-Aware Forecast and Evidence Head:* an evaluation layer that exposes direct, residual-corrected, and frozen Chronos-Bolt paths, transparent shift indicators, probabilistic diagnostics, and machine-readable forecast evidence under one chronological protocol.

The evaluation uses a final chronological holdout and includes linear, boosted tree, neural, retrieval-only, placebo-fusion, and frozen-TSFM baselines. Paired inference uses 168-hour blocks, 5,000 bootstrap resamples, HAC-corrected Diebold–Mariano tests, and Holm adjustment across the declared comparison family. This design makes negative evidence as visible as positive evidence and does not interpret event association as causation.

II. LITERATURE REVIEW AND RELATED WORK

A. STATISTICAL, MACHINE-LEARNING, AND DEEP FORECASTERS

Forecasting methods differ in how they trade inductive bias, data demand, and computational cost. Autoregressive and seasonal models remain informative baselines because their failure modes are easy to inspect. Tree ensembles such as XGBoost and LightGBM extend this setting to nonlinear lag, rolling, weather, and calendar interactions without requiring a sequence encoder [10], [11]. Recurrent LSTM and GRU models instead learn stateful temporal representations [12], [13]. Systematic studies nevertheless caution that recurrent networks are not automatic replacements for strong statistical baselines and require careful preprocessing and evaluation [14]. DeepAR extends the recurrent formulation to probabilistic forecasting across related series [15]. Large benchmark archives, including the Monash Forecasting Repository, further support reproducible comparison across heterogeneous time-series domains [16]. Transformer families emphasize longer dependencies: Informer reduces attention cost, Autoformer and FEDformer use decomposition, Crossformer and iTransformer model cross-variable structure, and TimesNet reorganizes temporal variation into two-dimensional patterns [17]–[22]. These gains do not imply that greater architectural complexity is always necessary. DLinear and PatchTST show that simple decomposition or patch-based inductive biases can remain highly competitive [23], [24], whereas TimeMixer uses explicit multiscale decomposition and mixing [25]. The evaluation therefore retains seasonal and boosted tree baselines instead of treating a deep backbone as the default winner. This choice is consistent with recent journal surveys, which emphasize that model selection should reflect horizon, covariate structure, uncertainty needs, and data scale rather than architecture novelty alone [26], [27].

B. TIME-SERIES FOUNDATION MODELS

Time-series foundation models (TSFMs) seek transfer across datasets rather than training a separate representation from scratch for each series. TimesFM uses a decoder-only forecasting design; Moirai trains a universal forecasting Transformer; MOMENT supports several time-series tasks; Chronos represents values as tokens; and Tiny Time Mixers emphasizes compact zero- and few-shot transfer [2]–[6]. Timer, Timer-XL, UniTime, and Time-LLM explore generative pretraining, long-context scaling, cross-domain unification, or language-model reprogramming [28]–[31]. Frozen pretrained models have also been evaluated as general computation engines for time-series analysis [32]. Collectively, these studies motivate zero-shot evaluation, but they also show that transfer performance depends on input representation and domain shift. The present study therefore compares frozen Chronos-Bolt against locally supervised models on identical forecast origins rather than assuming foundation-model superiority.

C. RETRIEVAL-AUGMENTED FORECASTING

Retrieval augmentation separates parametric knowledge from an external memory. In natural-language processing, RAG and Dense Passage Retrieval established the retrieve-then-condition pattern [7], [8]. In time series, retrieval has been studied through diffusion forecasting and assisted generation workflows [33], [34]. TimeRAF provides the closest methodological reference: it constructs a time-series knowledge base, trains an MLP dual-encoder retriever, selects top-k candidates, and integrates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Its ablations separate retrieval quality, integration, candidate count, knowledge-base composition, and backbone choice.

The present framework retains the external-memory principle but addresses a different experimental question. It uses transparent random, cosine, and event-context similarity instead of a learned dual encoder. Its principal TRACE-RAF path retrieves historical model residuals and applies a learned trust gate; a separate Chronos-Bolt arm evaluates output-space fusion rather than internal Channel Prompting. The implementation is therefore not a reproduction of TimeRAF. It is an audited environmental adaptation designed to test whether contextual analogues add information after a strong local forecast has been formed.

D. AIR-QUALITY FORECASTING AND OFFICIAL CONTEXT

PM2.5 forecasting commonly combines pollutant history with temperature, humidity, wind, precipitation, and pressure. Recent reviews describe a shift from single-station recurrent models toward attention-based, hybrid, and spatiotemporal systems [35], [36]. AirFormer, for example, separates deterministic spatiotemporal modeling from stochastic uncertainty estimation at nationwide scale [37]. Such multi-station designs address spatial dependence, whereas the present study examines a county-level aggregate and the provenance of contextual evidence. EPA AirData supplies regulated PM2.5 observations, NOAA GHCNh supplies hourly meteorology, and NOAA Storm Events supplies structured hazard records [38]–[40]. NOAA Hazard Mapping System products could add smoke evidence but were not included in the reported run [41].

E. DISTRIBUTION SHIFT AND FORECAST EVIDENCE

Temporal nonstationarity can alter means, variances, and relationships between inputs and future values. Adaptive normalization methods address this problem by estimating statistics at local temporal scales [42]; TRACE-RAF instead uses interpretable shift indicators to audit regimes without claiming that shift has been eliminated. Explanation methods also differ in scope. SHAP attributes predictions to model inputs [43], while information-theoretic saliency supports counterfactual inspection of probabilistic forecasts [44]. The implemented evidence layer is narrower: it records signed XGBoost effects, retrieved cases, event identifiers, drift components, and an uncertainty proxy. These records support traceability, but they do not establish causal effects of weather or events.

III. CRITICAL GAPS AND LIMITATIONS OF PREVIOUS STUDIES

Prior work leaves four connected gaps for event-sensitive urban air-quality forecasting. First, temporal forecasters and numerical retrieval systems do not necessarily preserve the provenance, coverage, and availability assumptions of local contextual records. A numerically similar PM2.5 window can occur under a different weather or event regime, and a retrospective event label can be mistaken for information available at forecast time. The SACB addresses this gap by joining official records under explicit source, coverage, and availability audits.

Second, historical retrieval can introduce look-ahead bias when a candidate input or target overlaps the query context. Similarity alone does not prevent a near-duplicate future from entering the retrieved evidence. The LSER addresses this gap through a strict temporal embargo and training-history restriction, followed by separately auditable PM2.5, weather, calendar, and event similarity terms.

Third, raw analogue futures can duplicate or conflict with information already captured by a strong supervised model. TRACE-RAF addresses this gap by storing strictly out-of-fold residual trajectories and learning an origin-level trust gate from validation-only correction utility. It therefore asks whether an analogue explains the remaining error rather than whether its raw future should replace the base forecast.

Fourth, retrieval gains are often reported without testing whether they persist under distribution shift, survive simple forecast-averaging placebos, or can be traced to stored evidence. The DFEH addresses this gap by evaluating direct, residual-corrected, and frozen Chronos-Bolt paths on common origins, exposing shift indicators, and retaining machine-readable feature, retrieval, event, and uncertainty evidence. This component supports comparison and auditability; it does not convert contextual association into causal explanation.

These mappings delimit the novelty of the study. TRACE-RAF does not contain a learned dual-encoder retriever or internal Channel Prompting, and the present single-county experiment does not establish geographic generalization.

IV. CORE CONTRIBUTIONS

The proposed framework contributes four mechanisms corresponding one-to-one with the gaps above:

- *Source-Audited Context Builder (SACB):* constructs an aligned 85-dimensional context from official PM2.5, meteorological, calendar, and event records while retaining source and availability metadata.
- *Leakage-Safe Event-Context Retriever (LSER):* retrieves training-history analogues only after enforcing a candidate-target embargo, then exposes the contribution of each similarity channel and the retrieved trajectory evidence.
- *Trust-gated Residual Analog Correction (TRACE):* constructs an embargoed memory of out-of-fold forecast residuals, aligns retrieved corrections to the query scale,

and applies a validation-trained reliability gate rather than uniform forecast replacement.

- *Drift-Aware Forecast and Evidence Head (DFEH):* places direct horizon models, TRACE-RAF, frozen-TSFM controls, drift-state analysis, and traceable forecast evidence within one chronological evaluation contract.

The empirical contribution is intentionally qualified. TRACE-RAF improves on its context ensemble in nominal MAE inference and remains competitive with the strongest nonlinear baseline, but the event channel itself has no verified all-origin effect and no favorable component comparison survives family-wise Holm correction.

V. PROBLEM FORMULATION

Let x_t denote the PM2.5 measurement at time t. For a forecast time t, the historical air-quality lookback window is

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t], \tag{1}$$

where L is the lookback length. Let W_t denote weather covariates aligned with the lookback window, C_t denote calendar features, and E_t denote event context available under the recorded source assumption at the forecast origin. The target associated with the input in (1) is

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \ldots, x_{t+H}], \tag{2}$$

where H is the forecasting horizon. Equation (2) therefore defines the 24-step vector evaluated at every valid origin.

The forecasting goal is to learn a function

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where R_t^{ts} is retrieved time-series knowledge, R_t^{ev} is retrieved event knowledge, and D_t contains distribution-shift indicators. The implemented target setting is $L = 168$ hours and $H = 24$ hours. Equation (3) states the full information interface; the implemented modules that construct these arguments are specified in the next section.

The evaluation uses a time-based split:

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

where training origins span January 8, 2019–August 23, 2024, validation origins span August 25, 2024–August 23, 2025, and test origins span August 25, 2025–December 30, 2025. These explicit cutoffs yield 45,784, 7,950, and 2,460 windows, respectively. The ordering in (4) is retained for feature fitting, retrieval eligibility, model selection, and final evaluation; the test partition is not used to preserve a nominal percentage.

Table 1 summarizes the essential notation.

TABLE 1. Main Symbols Used in the Proposed Formulation

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
E_t	Event context available at the forecast origin
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event knowledge
D_t	Drift indicators
q_t^B	Non-event base context in $\mathbb{R}^{46}$
q_t	SACB context vector in $\mathbb{R}^{85}$
G_t	LSER top-k retrieved set
$\rho(G_t)$	Retrieval summary in $\mathbb{R}^{51}$
$\hat{Y}_t^B$	Convex context-ensemble forecast
Δ_t	Retrieved residual correction in $\mathbb{R}^H$
z_t	TRACE reliability vector in $\mathbb{R}^{18}$
g_t	Validation-calibrated TRACE gate in $[0,1]$
$\mathcal{O}_t$	Machine-readable DFEH evidence record
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

VI. METHODOLOGY

TRACE-RAF is implemented through four coordinated modules. The Source-Audited Context Builder (SACB) converts heterogeneous official records into a chronological hourly context while retaining provenance and availability metadata. The Leakage-Safe Event-Context Retriever (LSER) constructs an embargoed historical memory and selects analogues according to PM2.5, weather, calendar, and event similarity. The TRACE core converts out-of-fold errors from a supervised context ensemble into an analogue residual memory and learns when those residuals are trustworthy. The Drift-Aware Forecast and Evidence Head (DFEH) exposes the final forecast, comparison paths, shift indicators, and a machine-readable evidence record. All modules operate on one-dimensional time series or tabular features; no image operator, internal TSFM Channel Prompting, or test-time parameter fitting is implied.

A. FRAMEWORK OVERVIEW AND FORWARD PASS

Figure 1 summarizes the complete dataflow. EPA PM2.5, NOAA GHCNh weather, calendar, and NOAA Storm Events records first enter the SACB, which audits, aligns, and transforms them into query contexts and input–target windows. The LSER applies the historical embargo before ranking eligible candidates. In parallel, expanding-window XGBoost and LightGBM models create strictly out-of-fold residuals for the training memory. TRACE-RAF retrieves and aligns those residuals, estimates their reliability, and adds a gated correction to the context ensemble. The DFEH records the forecast, retrieval cases, event identifiers, drift evidence, and uncertainty fields; frozen Chronos-Bolt remains a separately evaluated benchmark path.

For each forecast origin t, the complete mapping is

$$\hat{Y}_{t+1:t+H} = F_{\theta,\psi}\big(\phi(X_t, W_t, C_t, E_t), G_t, D_t, \mathcal{R}\big), \tag{5}$$

where $\phi(\cdot)$ denotes the tabular feature construction, G_t denotes retrieved historical evidence, D_t denotes drift evidence, and $\mathcal{R}$ is the out-of-fold residual memory. The key methodological constraint is that every candidate target and every residual in G_t and $\mathcal{R}$ ends before the query lookback begins. Equation (5) is the compact functional form that the tensor-level forward pass below expands.

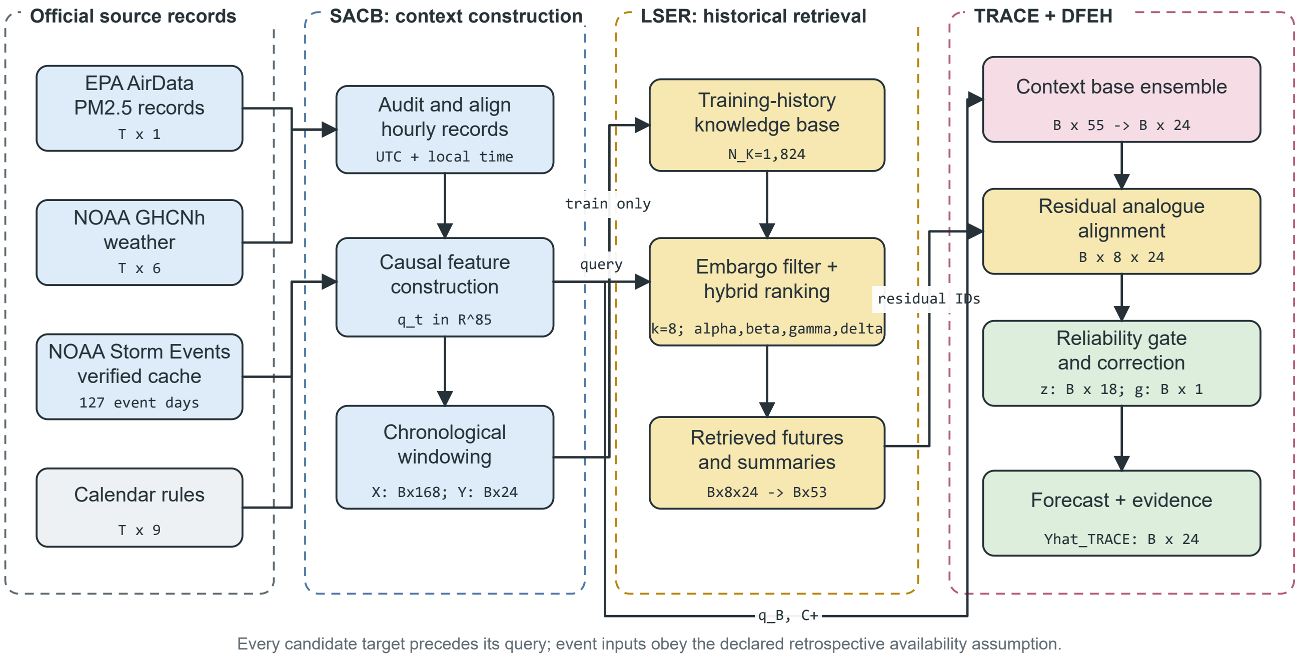

FIGURE 1. Proposed TRACE-RAF pipeline. Official records pass through SACB, training-history candidates are filtered and ranked by LSER, out-of-fold residual analogues are scaled by the TRACE trust gate, and DFEH produces the 24-hour forecast with traceable evidence. Tensor annotations show the manifest-backed configuration.

For a batch of B origins, the implemented forward pass can be expanded in tensor notation as

$$\begin{aligned}
Q^B &= \mathrm{SACB}_B(X, W, C) \in \mathbb{R}^{B\times 46},\\
Q &= [Q^B \,\|\, E] \in \mathbb{R}^{B\times 85},\\
G^r &= \mathrm{LSER}(Q, \mathcal{R}) \in \mathbb{R}^{B\times 8\times 24},\\
(\hat{Y}^L, \hat{Y}^X) &= \mathcal{H}(Q^B, C^+) \in (\mathbb{R}^{B\times 24})^2,\\
\hat{Y}^B &= \eta \hat{Y}^L + (1-\eta)\hat{Y}^X \in \mathbb{R}^{B\times 24},\\
(\Delta, Z) &= \mathcal{A}(G^r, \hat{Y}^B, D) \in \mathbb{R}^{B\times 24} \times \mathbb{R}^{B\times 18},\\
g &= \lambda\, \mathrm{clip}(h_\psi(Z), 0, 1) \in [0,1]^{B\times 1},\\
U &= \hat{Y}^B + g \odot \Delta \in \mathbb{R}^{B\times 24},\\
(\hat{Y}, \mathcal{O}) &= \mathrm{DFEH}(U, D, G^r) \in \mathbb{R}^{B\times 24} \times \mathcal{E}.
\end{aligned} \tag{6}$$

where $\eta = 0.6521$ is the out-of-fold-selected LightGBM weight, Δ is the aligned residual correction, Z contains 18 target-free reliability features, and $\lambda = 0.25$ is the validation-selected correction strength. The symbol $\mathcal{E}$ denotes the schema of evidence record $\mathcal{O}$. The operators $\mathcal{H}$ and $\mathcal{A}$ denote the paired context heads and residual-alignment stage, respectively. Here Q^B contains the 23 PM2.5, 14 weather, and nine calendar features; the 39 event features enter retrieval and gate evidence through Q but are excluded from both base heads. At each horizon, nine known future-calendar terms augment Q^B, giving 55 inputs to each base regressor. Equation (6) defines the proposed path; the feature-level M09 and frozen Chronos-Bolt arms are retained as controlled comparisons rather than components hidden inside TRACE-RAF.

B. SOURCE-AUDITED CONTEXT BUILDER

The SACB is motivated by the need to preserve data provenance and temporal availability before model fitting. As shown in Figure 2, source-specific checks precede feature construction. EPA monitor records are aggregated to an hourly county median, NOAA weather is aligned from the selected surface station, event archives are hash-verified against a manifest, and calendar values are derived from the local timestamp. The resulting context is then windowed and split chronologically; target values are never interpolated.

1) Dataset Construction

Hourly PM2.5 observations are taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [38]. The data audit selects one monitoring site using coverage and continuity criteria when a single site satisfies the coverage criterion. Because no individual monitor met this criterion, the target is the hourly county median across official Los Angeles County AQS monitors. Timestamps are converted to `America/Los_Angeles` with explicit daylight-saving handling. Short input gaps may be filled only from past-available observations; targets are never interpolated. Weather covariates come from the nearest suitable NOAA Global Historical Climatology Network hourly (GHCNh) station [39]. Calendar features include cyclic hour, weekday, and month terms, weekend status, US federal holidays, and California holidays.

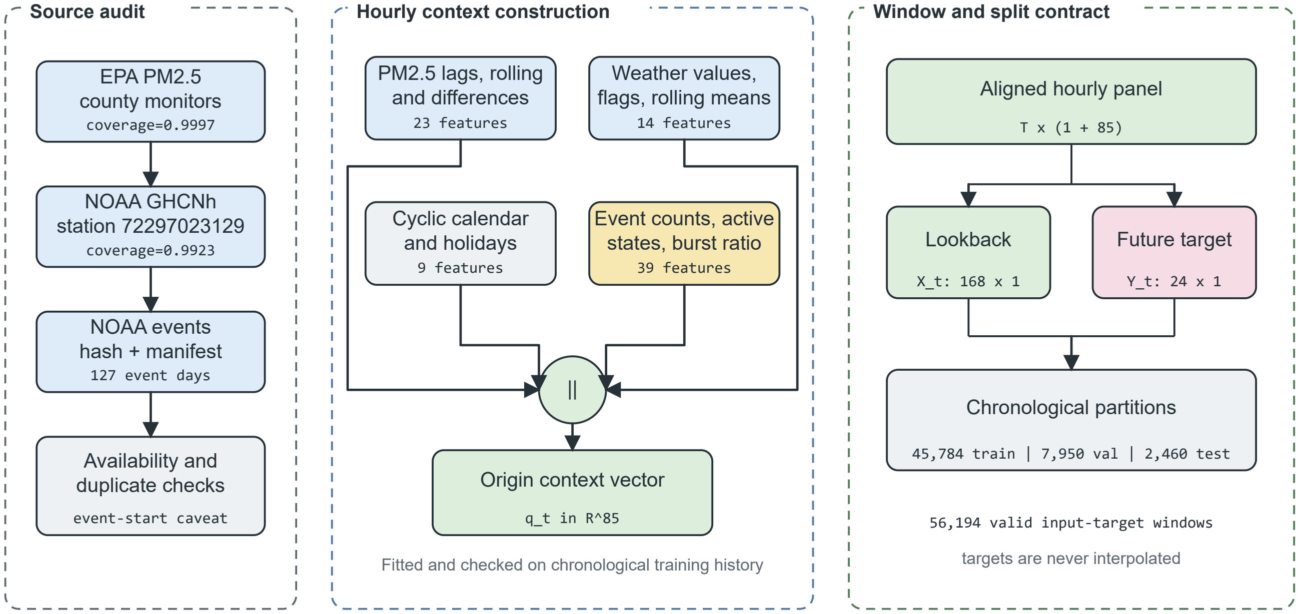

FIGURE 2. Source-Audited Context Builder. Source checks feed four feature groups that form an 85-dimensional origin vector. The aligned panel is converted into 168-hour inputs and 24-hour targets before the chronological split.

At each origin, the feature groups are concatenated as

$$\begin{aligned} q_t^B &= [p_t \parallel w_t \parallel c_t] \in \mathbb{R}^{23+14+9} = \mathbb{R}^{46}, \\ q_t &= [p_t \parallel w_t \parallel c_t \parallel e_t] \in \mathbb{R}^{23+14+9+39} = \mathbb{R}^{85}. \end{aligned} \tag{7}$$

where p_t, w_t, c_t, and e_t are respectively the PM2.5, weather, calendar, and event feature vectors. The PM2.5 group contains fixed lags, rolling moments and extrema, differences, the current value, and a fill flag. The weather group contains six measured variables, a precipitation-missing flag, derived condition flags, a stagnation proxy, and 24-hour rolling means. The event group contains one-hour and active counts, 24/72/168-hour counts, category-specific 72-hour counts, and an event-burst ratio. Thus, Equation (7) fixes both feature orderings: the 46-dimensional base context used by C01, M04, C05, and C06, and the 85-dimensional event-aware context used by LSER and event-conditioned controls.

All normalization statistics are fitted on the training split only. Query and candidate windows use the same preprocessing convention so that similarity is not confounded by inconsistent scaling. The reported run contains 56,194 valid windows with $X \in \mathbb{R}^{56194\times168}$ and $Y \in \mathbb{R}^{56194\times24}$. Of these, 45,784 belong to training, 7,950 to validation, and 2,460 to the final test holdout.

C. LEAKAGE-SAFE EVENT-CONTEXT RETRIEVER

The LSER separates temporal eligibility from relevance. Figure 3 first shows the embargo: a candidate is considered only if its complete target precedes the beginning of the query lookback. The eligible candidates are then compared along four independently stored similarity channels. The selected futures are aligned to the query scale and summarized for the forecasting head. This ordering prevents a high similarity score from overriding the temporal constraint.

1) Time-Series Knowledge Base

The time-series knowledge base contains historical PM2.5 windows. Each record stores a window identifier, start time, end time, input and future values, weather summary, calendar summary, normalized window vector, split, and normalization identifier. The primary knowledge base uses a 24-hour origin stride and contains 1,824 training windows. Sensitivity arms use 6-hour and 1-hour strides, yielding 7,608 and 45,784 candidates, respectively. Candidate overlap does not relax the query-specific embargo in Equation (9); it changes memory density only.

Formally, the knowledge base is

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^{N}, \tag{8}$$

where u_i is the candidate origin, X_i and Y_i are the candidate lookback and future windows, m_i is a weather summary, c_i is a calendar summary, and e_i is the event-context summary available at u_i. For a query origin t, the temporally eligible set is

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1,\ i \in \mathcal{T}_{train}\}, \tag{9}$$

for validation and test queries. The strict inequality in (9) ensures that a candidate's target period finishes before the query input period begins. The training-history condition in (9) is also enforced for final validation and test retrieval.

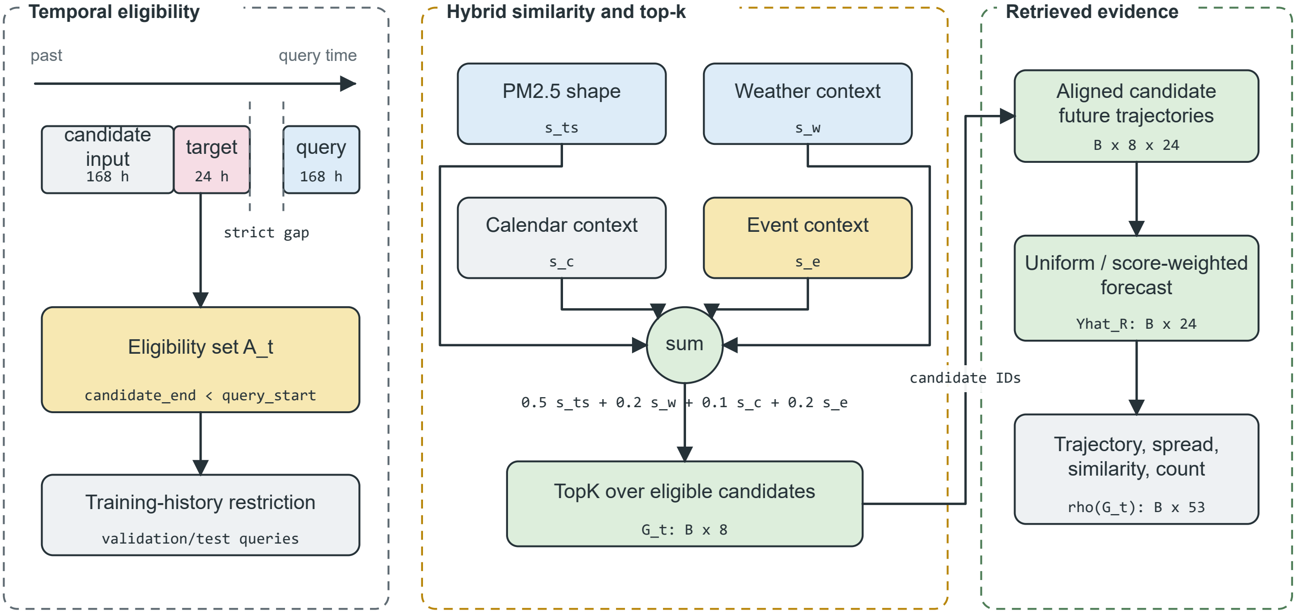

FIGURE 3. Leakage-Safe Event-Context Retriever. A strict candidate-target embargo and training-history restriction are applied before hybrid ranking. The top eight candidates provide aligned future trajectories and a 53-dimensional retrieval summary.

Equation (8) defines the stored candidate schema, while Equation (9) defines the admissible subset at query time.

Given a query window X_t, the time-series retriever returns

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \dots, r_{t,k}^{ts}\}, \tag{10}$$

where k is the number of retrieved candidates. The default is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$. Equation (10) denotes the returned records before their future trajectories are summarized.

2) Event Knowledge Base

The event knowledge base contains source-preserving records relevant to Los Angeles County. NOAA Storm Events is the required structured source, whereas NOAA Hazard Mapping System fire and smoke records remain an unmeasured optional extension [40], [41]. Stored fields include event identifier, start and end, information-availability assumption, location, category, source, source URL, title, and summary. The reported records cover wildfire, high wind, heavy rain, flood, and other weather categories; feature slots for absent configured categories remain zero.

Model inputs represent events with hourly counts, rolling 24-, 72-, and 168-hour counts, category-specific 72-hour counts, active-event counts, and an event-burst ratio. Recent source identifiers are retained as forecast evidence. Target-horizon overlap is used only as a post-hoc subset label and never as a model input. NOAA Storm Events does not provide a machine-readable publication timestamp, so the implementation records event start as a retrospective availability assumption. Strict operational experiments require records with genuine publication or issue times.

3) Hybrid Ranking and Future Alignment

Both query and candidate PM2.5 windows are standardized by their own means and standard deviations and normalized to unit length. Weather and calendar groups are standardized using candidate statistics and mapped cosine similarity to $[0, 1]$. Event similarity is clipped to $[0, 1]$ with explicit zero-vector handling. The hybrid retrieval score combines time-series similarity, weather relevance, calendar relevance, and event relevance:

$$s(q,i) = \alpha s_{ts}(q,i) + \beta s_w(q,i) + \gamma s_c(q,i) + \delta s_e(q,i), \tag{11}$$

where s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity, and s_e is event relevance. The implemented weights are $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$. These are the predeclared primary weights rather than a test-selected optimum; the executed event-weight sweep is reported in Section VIII-C. Equation (11) keeps each relevance channel explicit so that its contribution can be inspected without a learned latent scoring function.

The retrieved set is selected as

$$G_t = \mathrm{TopK}_{i \in \mathcal{A}_t}\ s(t,i). \tag{12}$$

Equation (12) is evaluated only after applying $\mathcal{A}_t$. Candidate futures are restored to the query scale as

$$\widetilde{Y}_{t,i} = \mu_t + \sigma_t \left(\frac{Y_i - \mu_i}{\max(\sigma_i, \epsilon)} \right), \qquad i \in G_t, \tag{13}$$

where (μ_t, σ_t) and (μ_i, σ_i) are the query and candidate look-back statistics. Retrieval-only forecasts average $\widetilde{Y}_{t,i}$ uniformly or by non-negative normalized scores. The feature-level path retains the mean trajectory, horizon-wise spread, mean and maximum time-series similarity, and candidate count, yielding $\rho(G_t) \in \mathbb{R}^{51}$. The affine transformation in Equation (13) preserves the candidate trajectory shape while restoring the query's local scale.

D. TRUST-GATED RESIDUAL ANALOG CORRECTION

TRACE-RAF changes what is retrieved. A raw candidate future contains both the part already predictable from current context and the part missed by the local model. Adding that future to a strong supervised forecast can therefore double count predictable structure. TRACE-RAF instead constructs a memory of out-of-fold residual trajectories and learns an origin-level trust coefficient for the retrieved correction. Figure 4 shows this residual path from the two context models through the residual memory, trust gate, final forecast, and evidence record.

1) Out-of-Fold Base Forecasts and Residual Memory

The base forecast is a convex combination of LightGBM and XGBoost context models,

$$\hat{Y}_t^B = \eta \hat{Y}_t^L + (1-\eta)\hat{Y}_t^X, \qquad 0 \le \eta \le 1, \tag{14}$$

where η is fitted only from expanding-window out-of-fold (OOF) training predictions. Two OOF prediction blocks cover fractions $[0.50, 0.75)$ and $[0.75, 1.00)$ of the training sequence. For each block, every fitting target ends before the first prediction input begins. The corresponding audit records 22,728 and 34,154 fit origins, 11,446 prediction origins per block, and a passed embargo in both cases.

For an OOF candidate i, the stored residual is

$$\varepsilon_i = Y_i - \hat{Y}_i^{B,\mathrm{OOF}} \in \mathbb{R}^H. \tag{15}$$

Residuals are clipped independently by horizon at their training-only 0.01 and 0.99 quantiles. The selected 24-hour residual-memory stride provides 909 valid candidates; validation also evaluates 3,795 and 22,892 residual candidates at 6-hour and 1-hour strides. The final stride is chosen before test retrieval.

2) Residual Retrieval and Scale Alignment

LSER ranks residual-memory candidates with the same temporal eligibility rule as Equation (9). If $a_{t,i}$ is the non-negative normalized retrieval weight, candidate residuals are transferred to the query scale and aggregated as

$$\Delta_{t,h} = \sum_{i \in G_t} a_{t,i} \frac{\sigma_t}{\max(\sigma_i, \epsilon)} \varepsilon_{i,h}, \qquad \sum_{i \in G_t} a_{t,i} = 1. \tag{16}$$

The same aligned residuals define a horizon-wise spread vector. In contrast to Equation (13), Equation (16) transfers a historical model error rather than a historical target level.

3) Reliability Gate

For each origin, an 18-dimensional target-free vector z_t summarizes the base-forecast mean, standard deviation, and maximum; correction mean, standard deviation, and maximum absolute magnitude; correction-spread mean and maximum; mean and maximum retrieval similarity; logarithmic selected and eligible candidate counts; selected-event fraction and whether event conditioning was applied; mean and maximum LightGBM–XGBoost disagreement; and the continuous and binary drift indicators. During validation only, the ideal scalar correction coefficient is

$$g_t^* = \mathrm{clip}\left(\frac{\langle Y_t - \hat{Y}_t^B, \Delta_t\rangle}{\|\Delta_t\|_2^2}, 0, 1\right). \tag{17}$$

A depth-2 histogram gradient-boosting regressor $h_\psi(z_t)$ learns this coefficient. The first 67% of validation origins fit the provisional gate; the remaining 33% select $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. The selected configuration is a 24-hour residual-memory stride and $\lambda = 0.25$; the gate is then refitted on the full validation partition without using test targets. The final forecast is

$$\hat{Y}_t^{\mathrm{TRACE}} = \hat{Y}_t^B + \lambda\, \mathrm{clip}(h_\psi(z_t), 0, 1)\Delta_t. \tag{18}$$

Equations (17) and (18) separate validation supervision from test-time inference: Y_t is required to train the gate but is absent from every test reliability feature.

E. DRIFT-AWARE FORECAST AND EVIDENCE HEAD

The DFEH is the common output and audit layer for the direct feature model, TRACE-RAF, the validation-only drift router, and the frozen Chronos-Bolt benchmark. As shown in Figure 4, the proposed path combines the context ensemble with a gated residual update. Separate branches retain M09 feature-level conditioning and Chronos-Bolt forecast fusion so their effects can be tested without conflating them with TRACE. Drift scores and flags are retained as reliability inputs, subset definitions, and evidence fields; the final TRACE gate is fixed before the test partition is opened.

1) Feature-Level Forecasting and Drift Diagnostics

For horizon h, the direct model input and prediction are

$$u_{t,h} = [q_t \,\|\, \rho(G_t) \,\|\, d_t \,\|\, c_{t+h}], \qquad \hat{y}_{t+h}^X = f_h(u_{t,h}), \tag{19}$$

where $u_{t,h} \in \mathbb{R}^{153}$ for the M09 comparison model and each f_h is an XGBoost regressor trained with squared-error objective. This design gives each forecast horizon a separate supervised function while retaining a common origin representation. Equation (19) accounts for the verified 153 features: 85 context, 53 retrieval, six drift, and nine future-calendar values. The drift score and binary flag support regime analysis and the evidence record; they do not adaptively select a forecaster in the reported experiment.

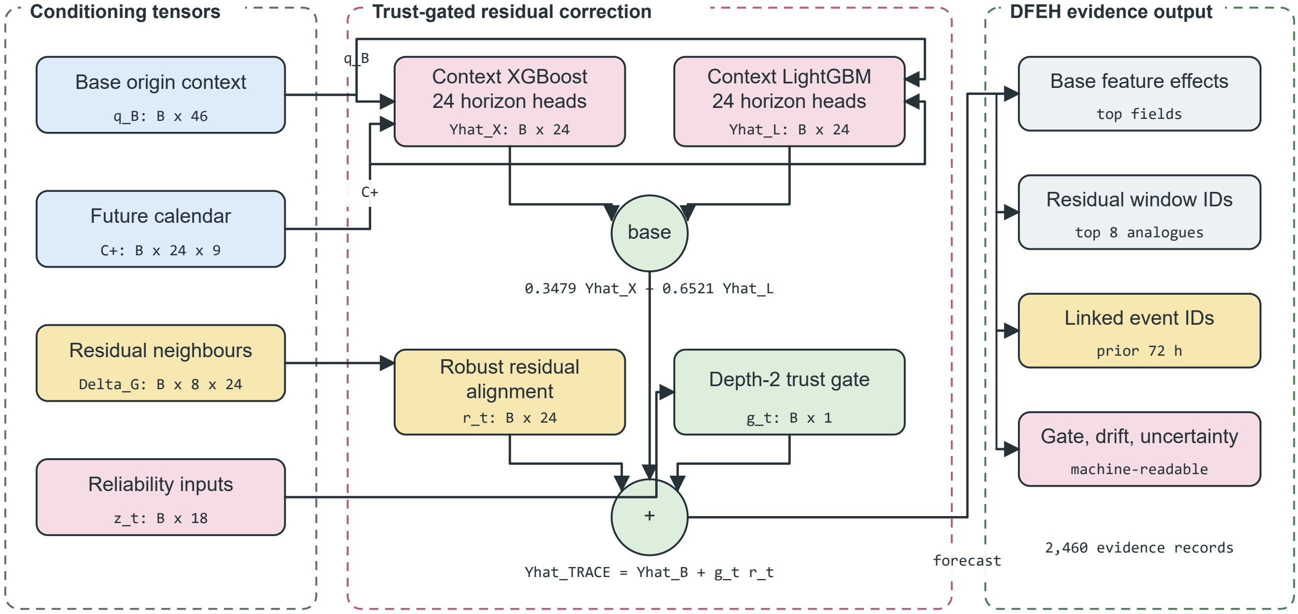

FIGURE 4. TRACE-RAF and the Drift-Aware Forecast and Evidence Head. Out-of-fold LightGBM and XGBoost predictions form a context ensemble, LSER returns aligned residual analogues, and an 18-variable reliability gate scales the correction. The final forecast is accompanied by feature effects, retrieved cases, event identifiers, drift components, and uncertainty evidence.

2) Frozen-TSFM Forecast Fusion

The Chronos-Bolt retrieval variant uses forecast-level fusion rather than internal Channel Prompting. Let $\hat{Y}_t^C$ be the frozen Chronos forecast and $\hat{Y}_t^R$ be the retrieved historical forecast. The fused prediction is

$$\hat{Y}_t^{CR}(\omega) = \omega \hat{Y}_t^C + (1-\omega)\hat{Y}_t^R, \tag{20}$$

where $\omega \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected $\omega = 0.25$. Because fusion occurs after both forecasts are produced, this branch does not modify Chronos-Bolt embeddings or weights. Equation (20) is therefore an output-space interpolation, not an internal TSFM prompting operation.

3) Distribution-Shift Diagnostics

The implementation represents distribution shift using transparent indicators rather than claiming verified concept-drift identification. The indicators are recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst ratio. The raw vector used by the code is

$$r_t = \left[|\mu_{24,t} - \mu_{168,t}|, \left|\log \frac{\sigma_{24,t}+\epsilon}{\sigma_{168,t}+\epsilon}\right|, 1 - \mathrm{clip}(\bar{s}_{ts,t}, 0, 1), \right.$$
$$\left. \sqrt{\frac{1}{4}\sum_{\ell=1}^{4}\left(\frac{w_{t,\ell} - \widetilde{w}_\ell^{tr}}{s_{w,\ell}^{tr}}\right)^2},\ \max(b_{event,t}, 0) \right], \tag{21}$$

where the weather term uses temperature, relative humidity, pressure, and wind speed standardized by training medians and standard deviations. Each component $r_{t,j}$ is then robustly scaled using the training median and median absolute deviation (MAD):

$$d_{t,j} = \frac{1}{6}\,\mathrm{clip}\left(\left| \frac{r_{t,j} - \mathrm{med}_{tr}(r_j)}{\max(1.4826\,\mathrm{MAD}_{tr}(r_j), \epsilon)} \right|, 0, 6 \right). \tag{22}$$

Finally, the continuous score and binary flag are

$$S_t = \frac{1}{5}\sum_{j=1}^{5} d_{t,j}, \qquad D_t = \mathbb{I}[S_t \geq Q_{0.90}(\{S_v : v \in \mathcal{T}_{val}\})]. \tag{23}$$

Equations (21)–(23) match the implemented detector: training data set the reference statistics, validation data set the 0.90-quantile threshold, and test data are transformed without refitting. The binary D_t is retained for diagnostic regime analysis and forecast evidence.

4) Evidence Record and Uncertainty Proxy

The evidence module uses signed local XGBoost contributions averaged across the 24 direct models, weather flags, the top three retrieved windows, recent event records, calendar context, forecast direction, and component-level drift evidence. Its uncertainty proxy is

$$u_t^{\mathrm{proxy}} = \sqrt{\bar{\sigma}_{R,t}^2 + \bar{e}_{val}^2}, \tag{24}$$

where $\bar{\sigma}_{R,t}$ is the mean horizon-wise spread of the retrieved trajectories and $\bar{e}_{val}$ is the validation residual RMSE. The result is a diagnostic scale, not a calibrated prediction interval.

Each evidence record is generated from stored fields rather than an unrestricted language model, and each event statement remains contextual rather than causal. Equation (24) is retained only as an evidence-field definition and is not used to construct confidence intervals.

F. RELATIONSHIP TO TIMERAF

TimeRAF retrieves top-k time-series windows using a learned MLP dual encoder and integrates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Table 2 identifies which principles are retained and which mechanisms differ. The comparison is methodological rather than a performance ranking because the studies use different datasets, horizons, and protocols.

VII. EXPERIMENTAL SETUP

This section describes the manifest-backed final experiment identified by `20260827T043457543402Z`. All model, retrieval-density, fusion, gate, and threshold choices were fixed from training and validation data before the final test partition was evaluated. The archived package contains the source, configuration, predictions, tables, evidence records, and figures used here.

A. DATASET DESCRIPTION AND AUDIT

The study period is 2019–2025. PM2.5 observations were taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [38]. No single monitoring site was used as the target series; the audited target is the hourly county median across official Los Angeles County AQS monitors. This fallback selected `06-037-COUNTY`, with 61,353 observed target hours, 191,284 contributing source records, and observed coverage of 0.9998. The PM2.5 unit was consistently recorded as micrograms per cubic meter.

Weather covariates were taken from NOAA GHCNh annual archives [39]. The selected station was `72297023129`, Long Beach / Daugherty Field / Airport, 10.36 km from the PM2.5 target centroid. The complete weather-feature coverage after allowed past-only short-gap filling was 0.9923, and the raw measured core coverage was 0.9882. Coverage exceeded 0.980 in every individual year and the record reached December 31, 2025.

Environmental events were loaded from hash-verified NOAA Storm Events annual detail archives [40]. Because direct access to NCEI was unavailable in the execution environment, the experiment used a locally staged cache containing unchanged NOAA annual archives and a source manifest with official URLs, byte counts, and SHA-256 hashes. The audit retained the delivery mode `verified_official_noaa_cache`. The event audit found 127 event days, 2,557 source-overlap days, and four qualifying event categories. The records include high-wind, flood, heavy-rain, other-weather, and wildfire categories. NOAA Storm Events does not expose machine-readable publication timestamps; therefore, event-aware outputs use the recorded event start time as a retrospective availability assumption. They are not strict real-time event forecasts. Table 3 reports the resulting readiness status. The source audit and window contract are visualized in Figure 2.

Table 4 consolidates the evaluated scope and the role of each source in the forecasting pipeline.

The origin-level attrition audit in Table 5 reconciles the hourly source panel with the model population. The largest reduction occurs when all causal origin features are required to be finite; 56,194 of 61,177 bounded candidate origins are retained (91.85%). All 24 target values for a retained origin are directly observed.

B. DATA AND CODE AVAILABILITY

The raw public records and download pages are available from EPA AirData, NOAA GHCNh, and NOAA Storm Events [38]–[40]. The experiment used a locally staged NOAA Storm Events cache because direct NCEI access was unavailable in the execution environment; every cached NOAA file was verified against the generated source manifest. Source code is available at https://github.com/shasabbir/event-time-raf. The implementation uses Python 3.12.13, NumPy, pandas, scikit-learn, XGBoost, PyTorch, and the `chronos-forecasting` package; the repository contains the configuration, notebooks, source modules, and tests needed to reproduce the pipeline when the public source files are accessible.

C. FORECASTING TASK AND SPLIT

The forecasting task predicts PM2.5 from one to 24 hours ahead using a 168-hour lookback window. Training contains 45,784 origins from January 8, 2019 through August 23, 2024; validation contains 7,950 origins from August 25, 2024 through August 23, 2025; and the frozen test holdout contains 2,460 origins from August 25 through December 30, 2025. Its targets end on December 31, 2025, producing 59,040 evaluated hourly points. Random splitting is not used, and test targets are not used for model or threshold selection.

D. MODELS

The baseline set spans seasonal rules, regularized linear regression, boosted trees, neural sequence models, retrieval-only forecasts, and a frozen TSFM. The proposed M13 TRACE-RAF model is compared with its C06 context ensemble and the event-free A03 gate. M12 is a separately validation-selected drift router, not a post-hoc artifact. Chronos placebos fuse the same frozen TSFM forecast with climatology and persistence, allowing forecast averaging to be separated from retrieval. The foundation-model evaluation uses `amazon/chronos-bolt-small`. Table 6 groups the evaluated configurations by their forecasting and retrieval roles.

E. IMPLEMENTATION AND HYPERPARAMETERS

The experiment was executed in a cloud-hosted Linux 6.12.90 environment using Python 3.12.13. The publication pipeline

TABLE 2. Methodological Comparison of TimeRAF and TRACE-RAF

Aspect	TimeRAF	TRACE-RAF
Primary objective	Improve zero-shot forecasting by augmenting a frozen TSFM with external time-series knowledge.	Evaluate source-audited, event-context residual retrieval for local PM2.5 forecasting under temporal shift.
Forecasting domain	Multi-domain benchmark datasets.	Los Angeles County hourly PM2.5, 2019–2025.
Knowledge base	Curated, consistently normalized time-series windows.	1,824 primary training windows, dense-memory sweeps up to 45,784 windows, and 909 selected OOF residual candidates.
Retriever	Learned MLP dual encoder with forecaster-guided training.	Random, cosine, and transparent hybrid similarity; no learned retriever.
Knowledge integration	Internal Channel Prompting of candidate embeddings.	Trust-gated residual correction; feature- and forecast-level fusion are comparison arms.
Backbone	Frozen TSFM, principally TTM-Base.	Convex LightGBM–XGBoost context ensemble; frozen `amazon/chronos-bolt-small` is a benchmark.
Leakage control	Non-overlapping knowledge windows and cross-dataset training constraints.	Candidate target must end before query lookback; validation and test retrieval use training history only.
Event handling	Not an explicit module.	Availability-tagged NOAA Storm Events features and source-record evidence, reported retrospectively.
Drift and evidence	Retrieved-case inspection.	Five-component drift audit, reliability-gated residuals, signed feature effects, retrieved IDs, event IDs, and uncertainty evidence.
Implementation scope	Learned retriever and Channel Prompting are implemented.	Those mechanisms are not implemented and remain future extensions.

TABLE 3. Data-Readiness Audit

Gate	Value	Status
PM2.5 observed coverage	0.9998	Pass
Weather complete coverage	0.9923	Pass
Event days	127	Pass
Event overlap days	2,557	Pass
Qualifying event categories	4	Pass
Strict event availability	False	Caveat

recorded 7,432.7 seconds of internal runtime. Key package versions recorded in the run manifest include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, LightGBM 4.6.0, PyTorch 2.10.0+cu128, Transformers 4.57.6, and chronos-forecasting 2.3.1. Table 7 reports the principal hyperparameters.

F. EVALUATION METRICS AND STATISTICAL CHECKS

The primary metrics are MSE, MAE, RMSE, MAPE, sMAPE, and R^2. MSE is retained as the primary squared-error criterion, while MAE and RMSE provide more interpretable PM2.5 error scales. For n observed forecast values y_j and predictions $\hat{y}_j$, the principal error metrics are

$$\begin{aligned} \mathrm{MSE} &= \frac{1}{n}\sum_{j=1}^{n}(y_j-\hat{y}_j)^2, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{j=1}^{n}|y_j-\hat{y}_j|, \\ \mathrm{RMSE} &= \sqrt{\mathrm{MSE}}, \end{aligned} \tag{25}$$

and explained variance is summarized by

$$R^2 = 1-\frac{\sum_{j=1}^{n}(y_j-\hat{y}_j)^2}{\sum_{j=1}^{n}(y_j-\bar{y})^2}. \tag{26}$$

Equation (25) is aggregated across all forecast origins and horizons for the overall result, while (26) uses the same set of points. MAPE and sMAPE are retained in machine-readable outputs but are not used for the main ranking because low PM2.5 values can make percentage errors unstable. Subset metrics are reported only when at least 50 forecast origins are available. The 72-hour event-context subset contains 70 origins and the drift subset contains 407; the 33 target-event and eight active-event origins are retained in counts but suppressed from performance tables. Pairwise comparisons use 5,000 paired moving-block bootstrap resamples with 168-hour blocks and Diebold–Mariano tests with 168 HAC lags. Both raw and Holm-adjusted p-values are archived. AQI threshold decisions are evaluated at 35.4 and 55.4 μg/m^3, while Chronos quantiles are evaluated by interval coverage, an approximate quantile-grid CRPS, and calibration error.

G. COMPARISON WITH EXISTING STUDIES

Direct numerical comparison with existing TSFM and air-quality papers is not valid because they use different datasets, forecast horizons, variables, and evaluation protocols. Table 8 therefore compares methodological coverage rather than claiming cross-paper performance dominance.

VIII. RESULTS AND DISCUSSION

A. OVERALL COMPARISON WITH BASELINES

Table 9 reports the final-holdout results. Ridge regression with context features is strongest overall, with MSE 39.144, MAE 4.079, RMSE 6.257, and $R^2 = 0.457$. TRACE-RAF obtains MSE 40.062, MAE 4.134, and RMSE 6.329. It reduces MSE by 4.63% relative to M04 XGBoost context and is within 0.07% of C05 LightGBM context, but it does not exceed Ridge or LSTM. This ranking is important: the proposed correction is competitive with nonlinear context models, while the smooth county-median target and late-2025

TABLE 4. Dataset and Source Summary

Data group	Source	Evaluated scope	Use in pipeline
PM2.5	EPA AirData/AQS hourly parameter 88101	Los Angeles County, 2019–2025; 61,353 observed target hours	Target sequence and lag/rolling features
Weather	NOAA GHCNh, station `72297023129`	2019–2025; 0.9923 usable core coverage	Temperature, humidity, wind, pressure, and rain features
Events	NOAA Storm Events annual detail archives	127 event days; four qualifying categories	Retrospective event-start features and forecast evidence
Calendar	Generated from timestamps and public holiday rules	2019–2025 hourly calendar	Hour, weekday, weekend, month, season, and holiday features

TABLE 5. Forecast-Origin Attrition Audit

Stage	Origins	Retained (%)
Bounded candidate origins	61,177	100.00
Chronological split eligible	61,129	99.92
Complete 168-hour lookback	61,121	99.91
Complete observed target	60,953	99.63
Complete causal origin features	56,194	91.85
Final retained windows	56,194	91.85

holdout remain favorable to regularized linear structure.

The frozen Chronos-Bolt evaluation completed successfully and validation selected a TSFM weight of 0.25. Event-retrieval fusion increases MSE from 47.534 to 48.609 and MAE from 4.202 to 4.574. The MAE increase of 0.372 remains significant after Holm adjustment (adjusted bootstrap $p = 0.046$; adjusted DM $p = 0.035$). Moreover, the persistence-fusion placebo reaches MSE 46.964, lower than both M10 and M11. Thus this run provides evidence against attributing a generic averaging effect to event retrieval. Chronos native quantiles remain useful diagnostically: M10 attains approximate CRPS 3.187, mean absolute calibration error 0.017, and 0.761 empirical coverage for the nominal 0.80 interval, whereas M11 degrades those values to 4.120, 0.176, and 0.206.

B. ABLATION STUDY

Table 10 isolates the residual-memory hypothesis from its integration rule. Every selectable value was chosen on the final 2,624 origins of the validation partition, beginning on May 3, 2025; the test partition was not used for selection. Raw-future fusion (A04) and the global residual scale (A06) both select zero and therefore reduce exactly to C06. This identity is a validation outcome rather than an implementation failure. The fixed, ungated residual update (A05) records MSE 44.959, whereas the learned origin-specific gate selects strength 0.25 and yields MSE 40.062. Thus, the ablation supports the gate's role as a conservative control mechanism, but not a claim that residual retrieval is uniformly beneficial.

Table 11 reports the declared paired MSE contrasts; negative differences favor the first model. Cosine retrieval features improve M04 nominally, and raw event features outperform the matched feature-count placebo nominally. TRACE-RAF has almost identical MSE to both its base ensemble and LightGBM. Its MAE differences are -0.021 versus C06 (unadjusted 95% block-bootstrap CI $[-0.038, -0.005]$) and -0.017 versus C05 (unadjusted 95% block-bootstrap CI $[-0.031, -0.001]$). Their Holm-adjusted bootstrap p-values are 0.355 and 0.490, respectively; the corresponding Holm-adjusted DM p-values are 0.381 and 0.516. The M13–A03 contrast is effectively zero, directly showing that event conditioning does not explain TRACE-RAF's aggregate result. Across the complete 29-comparison family for each metric, only the adverse M11–M10 MAE contrast remains significant after Holm correction.

The ablation does not form a monotone accuracy progression. PM2.5 history, weather, and calendar features dominate validation faithfulness; event, retrieval, and drift groups make smaller contributions. The resulting paper therefore attributes TRACE-RAF's behavior to selective residual correction, not to a demonstrated general event effect.

The k sensitivity results in Table 12 show that the retrieval-only cosine baseline improves as more neighbors are used within the tested range. The best retrieval-only configuration among $k \in \{1, 4, 8, 16\}$ is $k = 16$, with MSE 51.316. The reported feature models used the pre-specified default $k = 8$.

Knowledge-base density has a non-monotone effect. Table 13 shows that the 6-hour memory is strongest for both retrieval-only and feature-level event-aware variants; simply admitting every hourly candidate is not optimal. At 6-hour stride, event-conditioned retrieval improves on cosine retrieval by 1.669 MSE, whereas it is worse at the 24- and 1-hour strides. This interaction motivates reliability gating rather than a claim that event similarity is uniformly beneficial.

C. EVENT-SIMILARITY WEIGHT SENSITIVITY

The primary LSER event coefficient $\delta = 0.2$ was fixed in the configuration before final-test evaluation. Table 14 reports the complete executed sweep while holding the time-series, weather, and calendar coefficients fixed. The learned M08 validation MSE varies only from 35.081 to 35.278, and the event term changes the selected retrieval candidate for fewer than 2.9% of test queries relative to $\delta = 0$. The retrieval-only event-context MSE is lowest at the primary value, whereas the learned M08 model is numerically lowest at $\delta = 1.0$ on both validation and test. The fixed primary value is therefore not claimed to be optimal. More importantly, the sweep does not

TABLE 6. Evaluated Model Families

ID	Description
C00–C06	Hour–month climatology; Ridge and LightGBM context; calendar and event retrieval; raw-event XGBoost; convex context ensemble
M00–M02	Persistence, daily seasonal, and weekly seasonal naive baselines
M03–M09	PM2.5/context XGBoost; retrieval-only controls; feature-level retrieval, event, and drift variants
B00–B02	DLinear, PatchTST, and LSTM trained on the same PM2.5 windows
M10–M12	Frozen Chronos-Bolt, Chronos retrieval fusion, and validation-only drift router
M13	Proposed TRACE-RAF trust-gated residual analogue correction
A00–A03	Event-free, random-retrieval, matched feature-count placebo, and event-free TRACE ablations
P00–P01	Chronos fused with climatology or persistence placebos

TABLE 7. Experimental Hyperparameters and Runtime Settings

Component	Setting
Study period	2019–2025
Lookback and horizon	$L = 168$ hours, $H = 24$ hours
Split	45,784 train; 7,950 validation; 2,460 final-test origins
SACB feature groups	23 PM2.5, 14 weather, 9 calendar, 39 event features
SACB missing-data rule	Past-only filling for gaps up to 3 hours; no target interpolation
LSER retrieval	$k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$
LSER knowledge base	1,824 windows at 24-hour primary stride; 6/1-hour sweeps
LSER similarity weights	time series 0.5, weather 0.2, calendar 0.1, event 0.2 (predeclared primary)
DFEH direct input	153 features for each of 24 horizon regressors
DFEH XGBoost	350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$
TRACE base ensemble	LightGBM weight 0.6521; XGBoost weight 0.3479
TRACE residual memory	OOF boundaries 0.50/0.75/1.00; 909 selected candidates
TRACE trust gate	18 inputs; depth 2; 100 iterations; learning rate 0.05; selected strength 0.25
DFEH drift detector	5 components; 24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold
DFEH Chronos path	`amazon/chronos-bolt-small`, batch size 64
DFEH fusion weights	$\{0, 0.25, 0.5, 0.75, 1.0\}$; selected $\omega = 0.25$
Neural optimization	AdamW; seed 42; batch 256; at most 30 epochs; patience 5; learning rate 0.001; weight decay 0.0001
DLinear	moving-average kernel 25; selected epoch 6
LSTM	hidden size 64; 2 layers; dropout 0.1; selected epoch 2
PatchTST	patch/stride 24/12; 3 layers; $d = 64$; 4 heads; FFN 128; selected epoch 4
Forecast evidence record	Top 3 feature effects, windows, and recent event IDs
Inference	5,000 paired resamples; 168-hour blocks/HAC lags; Holm correction

TABLE 8. Methodological Comparison with Related Forecasting Work

Method family	TSFM	Retrieval	Events	Evidence	Comment
PatchTST-style Transformers [24]	No	No	No	Limited	Strong sequence modeling, but event context is not central.
Chronos [5]	Yes	No	No	Limited	Frozen zero-shot baseline used in this study.
TimeRAF [9]	Yes	Yes	No	Retrieval cases	Base retrieval-augmented TSFM method; does not target official event-context PM2.5.
XGBoost context baseline [10]	No	No	No	Feature importance	Strong supervised tabular baseline for local PM2.5.
TRACE-RAF	No[a]	Yes	Yes	Evidence-grounded	Adds official-source audit, OOF residual retrieval, a trust gate, drift diagnostics, and stored evidence.

[a]Chronos-Bolt is evaluated as a frozen benchmark, but the proposed TRACE-RAF forecast is a supervised residual-correction model rather than a foundation model.

reveal a stable event-specific accuracy advantage and leaves the paper's qualified event conclusion unchanged.

D. EVENT AND DRIFT SUBSETS

Table 15 reports the two pre-specified subsets that meet the 50-origin gate. The broader 72-hour event-context subset contains 70 origins; direct target-event overlap contains only 33 origins and is therefore not scored. Event-context targets have mean 6.26 and variance 11.26, compared with mean 11.14 and variance 73.20 outside that subset. Consequently, cross-subset MSE magnitudes are not interpreted as event difficulty. Within the same event-context origins, TRACE-RAF and its event-free ablation are indistinguishable. The drift subset contains 407 origins selected by the validation-calibrated DFEH threshold and exhibits much larger target variance (105.34), explaining part of its larger absolute error.

TABLE 9. Manifest-Backed Overall Test Results for 24-Hour PM2.5 Forecasting

Model	MSE	MAE	RMSE	R^2
C00 hour–month climatology	73.915	6.398	8.597	-0.025
M00 persistence	57.327	4.840	7.571	0.205
M03 XGBoost PM2.5	44.547	4.317	6.674	0.382
M04 XGBoost context	42.006	4.233	6.481	0.417
C01 Ridge context	**39.144**	**4.079**	**6.257**	**0.457**
C05 LightGBM context	40.089	4.151	6.332	0.444
B00 DLinear	40.305	4.097	6.349	0.441
B01 PatchTST	40.766	4.111	6.385	0.435
B02 LSTM	39.978	4.095	6.323	0.445
C06 context ensemble	40.115	4.156	6.334	0.444
M09 event-feature XGBoost	41.193	4.148	6.418	0.429
M13 TRACE-RAF	*40.062*	*4.134*	*6.329*	*0.444*
A03 TRACE-RAF without events	40.062	4.135	6.329	0.444
M10 frozen Chronos-Bolt	47.534	4.202	6.894	0.341
M11 Chronos + event retrieval	48.609	4.574	6.972	0.326
M12 validation drift router	41.466	4.190	6.439	0.425

TABLE 10. Validation-Selected TRACE-RAF Mechanism Ladder on the Final Holdout

ID	Forecast integration	Retrieval target	Integration control	MSE	MAE
C06	Base ensemble	None	None	40.115	4.156
A04	Base + raw analogue	Raw future	Weight 0.00 (validation)	40.115	4.156
A05	Base + residual analogue	OOF residual	Strength 1.00 (fixed)	44.959	4.250
A06	Base + residual analogue	OOF residual	Strength 0.00 (validation)	40.115	4.156
M13	Base + gated residual analogue	OOF residual	Gate strength 0.25 (validation)	**40.062**	**4.134**

TABLE 11. Selected 168-Hour Paired Block-Bootstrap Ablations

Comparison	MSE diff.	Unadj. 95% CI low	Unadj. 95% CI high	Holm-adjusted p
M04 − M03: weather/calendar	-2.542	-6.695	0.940	1.000
M07 − M04: cosine retrieval features	-1.071	-2.280	-0.041	1.000
M08 − M07: event-conditioned ranking	0.151	-0.447	0.789	1.000
C04 − A02: event signal/placebo	-0.524	-0.983	-0.142	0.389
M09 − M08: drift features	0.107	-0.386	0.502	1.000
M13 − C06: TRACE residual gate	-0.052	-0.393	0.353	1.000
M13 − A03: TRACE event channel	-0.0001	-0.006	0.008	1.000
M13 − C05: proposed/best tree	-0.027	-0.350	0.353	1.000
M11 − M10: Chronos retrieval	1.075	-7.177	6.712	1.000
M12 − M04: drift router	-0.540	-1.112	-0.066	0.747

TABLE 12. Cosine Retrieval Sensitivity to k

k	MSE	MAE	RMSE
1	84.602	5.868	9.198
4	57.092	5.011	7.556
8	53.549	4.860	7.318
16	51.316	4.773	7.164

The composite detector flags 407 test origins at threshold 0.2132, while a validation-calibrated two-sample KS benchmark flags 348 at threshold 0.4523. Their Jaccard agreement is only 0.117. The composite flag is therefore treated as a model-specific regime label, not a ground-truth definition of drift. The separately selected M12 router improves M04 nominally but does not survive Holm correction, reinforcing this cautious interpretation.

E. SITE-LEVEL AND DECISION SENSITIVITY

The county median is the primary target because no individual monitor reaches the 0.70 coverage gate. To test whether aggregation alone determines the event finding, three monitor-level arms were trained without changing their temporal protocol. Table 16 shows heterogeneous effects: event retrieval is worse at site 1103 and modestly better at sites 4009 and 9035. Event-only site subsets contain 10–33 origins and are not scored. These are internal target-sensitivity checks, not external geographic validation.

At the 35.4 μg/m^3 24-hour-mean threshold, only 37 of 2,460 test origins are exceedances; no origin exceeds 55.4 μg/m^3. Physical-threshold forecasts have very low recall. With validation-calibrated decision thresholds, LightGBM attains precision 0.234, recall 0.405, and F1 0.297; TRACE-RAF attains 0.213, 0.351, and 0.265. These sparse-event results are reported to connect the regression objective to air-quality decisions, but they are insufficient for a standalone

TABLE 13. Knowledge-Base Density Sensitivity on the Final Holdout

Stride (h)	Candidates	Cosine MSE	Event-retrieval MSE	M08 MSE	M09 MSE
24	1,824	53.549	54.098	41.085	41.193
6	7,608	**49.524**	**47.855**	**40.520**	**40.886**
1	45,784	52.738	54.253	41.172	41.749

TABLE 14. Sensitivity to the LSER Event-Similarity Weight; the Primary Configuration is Marked and Test Values Were Not Used for Selection

Event weight δ	M08 validation MSE	M08 test MSE	Event-context retrieval MSE	Candidate change vs. $\delta = 0$ (%)
0.0	35.227	40.842	12.487	0.00
0.2 (primary)	**35.201**	**41.085**	**11.961**	**2.70**
0.5	35.278	40.690	12.151	2.78
0.8	35.110	40.858	13.735	2.79
1.0	35.081	40.620	14.322	2.83

TABLE 15. Representative Event and Drift Subset Results

Subset	Model	Origins	MSE	MAE
Event context	C01 Ridge context	70	5.624	1.925
Event context	C06 context ensemble	70	6.568	2.101
Event context	M13 TRACE-RAF	70	6.552	2.097
Event context	A03 TRACE without events	70	6.553	2.095
Non-drift	C01 Ridge context	2,053	38.808	3.989
Non-drift	M13 TRACE-RAF	2,053	38.085	3.955
Drift	C01 Ridge context	407	40.843	4.536
Drift	C06 context ensemble	407	50.267	5.077
Drift	M13 TRACE-RAF	407	50.033	5.039
Drift	A03 TRACE without events	407	50.047	5.039

TABLE 16. Monitor-Level Event-Retrieval Sensitivity

Site	Coverage	Origins	M04 MSE	M08 MSE
06-037-1103	0.558	1,631	68.281	69.898
06-037-4009	0.510	1,732	73.506	73.095
06-037-9035	0.374	547	17.682	17.371

alerting claim.

F. HORIZON-WISE AND RETRIEVAL DIAGNOSTICS

Fig. 5 and Fig. 6 show horizon-wise test errors for principal supervised, feature-retrieval, frozen-TSFM, and TRACE-RAF comparators. The ranking changes with lead time, but no model exhibits an isolated late-horizon failure that would invalidate the aggregate comparison. Fig. 7 places the DFEH score and its fixed threshold on the test timeline. Figures 8 and 9 provide ordinary and event-context examples; they are qualitative checks and are not used to select models. All plots are regenerated from run `20260827T043457543402Z`.

G. FORECAST EVIDENCE AND AUDITABILITY ANALYSIS

Within the DFEH, a source-grounded forecast evidence record is retained rather than claiming a complete causal explanation. For each of the 2,460 test origins, the pipeline stores the run and window identifiers, origin time, retrieved historical windows, linked event records, leading feature effects, drift components, retrieval spread, validation residual

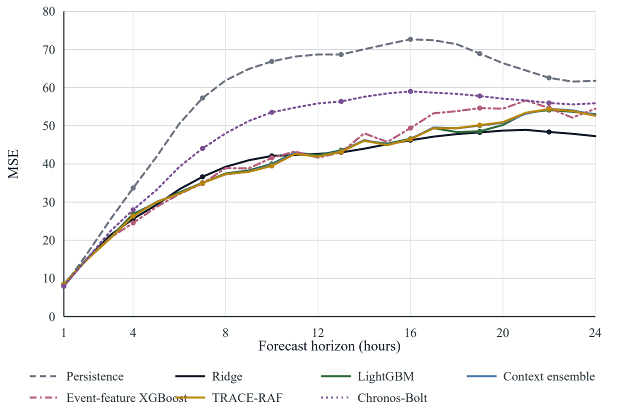

FIGURE 5. MSE by forecast horizon on the Los Angeles County test set.

RMSE, uncertainty proxy, drift flag, and event-availability mode. The resulting `explanations.parquet` file therefore has a one-to-one relationship with the evaluated forecasts. Its dominant local fields are usually current PM2.5 and recent rolling concentrations, while analogue and event identifiers expose the evidence available to the correction mechanism.

Table 18 summarizes the evidence layer. This design supports auditability because every evidence statement can be traced back to a model feature, a retrieved time-series win-

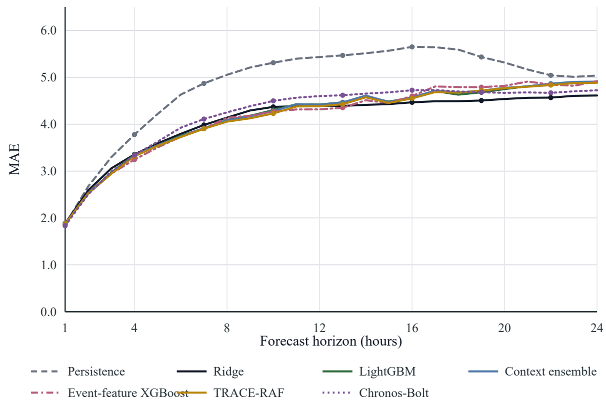

FIGURE 6. MAE by forecast horizon on the Los Angeles County test set.

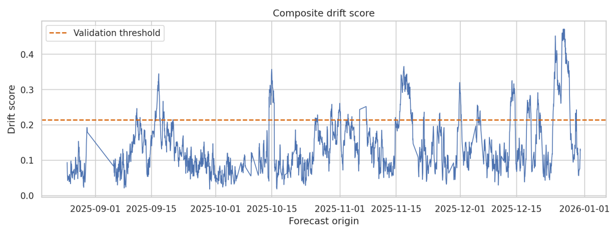

FIGURE 7. Composite drift score and the validation-calibrated threshold on the final holdout. The flag defines a diagnostic regime rather than verified distribution-shift ground truth.

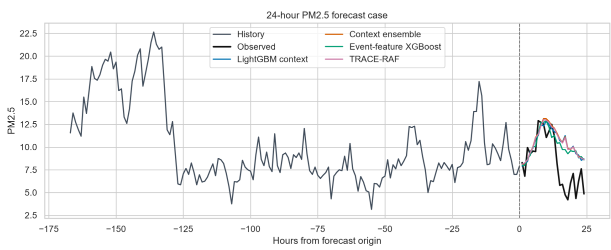

FIGURE 8. Representative 24-hour forecast from the final holdout. Model traces are plotted against the observed hourly target without post-test adjustment.

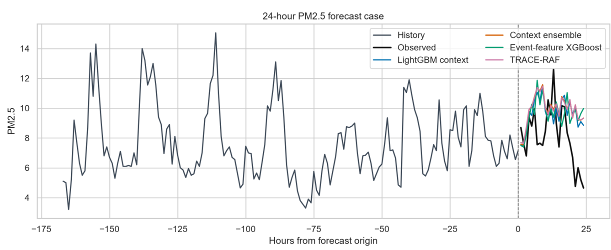

FIGURE 9. Representative forecast whose target lies within the declared 72-hour event-context window. The event designation is retrospective because Storm Events files do not expose publication timestamps.

dow, an official event record, or a drift component. It does not prove causality: event evidence is reported as contextual support, and the drift flag is treated as an uncertainty warning. Faithfulness is checked by perturbing complete feature families in the fitted M09 model. As shown in Table 19, PM2.5 history has by far the largest effect and weather and calendar information remain material. Event and retrieval permutations increase MSE, but their median-deletion effects are negative. These mixed signs prohibit a stronger claim that every stored evidence type is a stable causal explanation.

The lossless prediction archive also permits direct reconstruction of the final TRACE update, $\hat{Y}^{\mathrm{TRACE}} = \hat{Y}^{B} + g\Delta$. Here, g is the effective origin-level coefficient after applying the validation-selected strength of 0.25. Table 17 quantifies each component. Across all test origins, the learned gate reduces a mean absolute raw residual analogue of 1.708 to an applied correction of only 0.149 μg/m^3. Event-context origins receive a still smaller mean applied correction of 0.060 μg/m^3. These values expose how the final forecast was formed; they do not assign a causal contribution to the event records.

TABLE 17. Lossless TRACE Forecast-Contribution Decomposition

Subset	Origins	Mean g	Mean $\|\Delta\|$	Mean $\|g\Delta\|$
All	2,460	0.090	1.708	0.149
Event context	70	0.082	0.736	0.060
Non-event context	2,390	0.090	1.736	0.152
Distribution shift	407	0.076	2.386	0.175
No distribution shift	2,053	0.093	1.573	0.144

TABLE 18. Forecast Evidence Stored by TRACE-RAF

Evidence field	Interpretation role
Top feature effects	Local model drivers from the direct forecaster.
Retrieved windows	Historical analogues used by the retrieval module.
Event evidence IDs	Official storm-event context linked to the origin.
Drift components	Mean, variance, weather, similarity, and event-burst shift signals.
Uncertainty proxy	Combined retrieval spread and validation residual error.

TABLE 19. M09 Feature-Family Faithfulness Perturbations

Feature family	Permutation ΔMSE	Median deletion ΔMSE
PM2.5 history	25.575	13.531
Weather	3.440	3.089
Calendar	3.548	2.481
Events	1.162	-0.442
Retrieval	0.964	-0.551
Drift	0.268	0.125

H. COMPUTATIONAL COST AND REPRODUCIBILITY

The publication pipeline recorded 7,432.7 seconds of internal runtime in a Linux environment using Python 3.12.13. All three notebooks completed without cell errors. The recorded versions include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, LightGBM 4.6.0, PyTorch 2.10.0+cu128, Transformers 4.57.6, and

TABLE 20. Computational and Reproducibility Summary

Item	Recorded value
Run identifier	`20260827T043457543402Z`
Pipeline runtime	7,432.7 seconds
Platform	Linux 6.12.90
Python	3.12.13
Forecast horizon	24 hours
Lookback window	168 hours
Bootstrap resamples	5,000 in 168-hour blocks
Multiple testing	Holm adjustment over the comparison family
Chronos checkpoint	`amazon/chronos-bolt-small`
Code availability	https://github.com/shasabbir/event-time-raf

`chronos-forecasting` 2.3.1. The archive contains 43 result tables with a single run identifier, configuration and artifact hashes, prediction-level outputs, evidence records, and an independent claim-verification notebook. The TRACE subset-inference reconstruction agrees with stored predictions to a maximum absolute error of 9.85×10^{-17}. Hardware identifiers and a Git commit were not available inside the extracted notebook environment, so runtime is not interpreted as a hardware-normalized efficiency result.

The main interpretation is that supervised context models are unusually strong for this Los Angeles County task. TRACE-RAF improves on context XGBoost and is nominally better than the LightGBM comparator in both MSE and MAE, but Ridge remains best overall and no favorable TRACE contrast survives family-wise adjustment. The mechanism ladder further shows that validation rejects raw and globally scaled analogue fusion, while the learned gate applies only a small fraction of the retrieved residual. The nearly identical event-free ablation shows that the aggregate TRACE result comes from conservative residual correction, not measurable event conditioning. By contrast, adding event retrieval to frozen Chronos-Bolt significantly worsens MAE after Holm correction. The result is thus a reproducible negative and positive boundary: trust-gated residual analogues are competitive, whereas event-conditioned output-space fusion with frozen Chronos-Bolt-small is not supported in this holdout. This result does not test internal TSFM prompting or retrieval-augmented TSFMs generally.

A pre-specified January 2025 stress analysis contains 622 origins but belongs to the validation period and is reported only as development evidence. Ridge, Chronos-Bolt, the context ensemble, and TRACE-RAF obtain MSE 118.300, 116.012, 123.305, and 123.643, respectively. It is not an additional test set and does not alter the final model ranking.

Finally, all event-aware interpretations must be read with the availability caveat. NOAA Storm Events records are official and hash-verified, but the source does not provide machine-readable publication timestamps. The experiment records event start as the availability time, so event-related results are retrospective sensitivity results. A strict operational study would require event or smoke sources with genuine issue or publication times, such as an audited NOAA HMS cache or alert product archive.

IX. LIMITATIONS

The first limitation is event availability. NOAA Storm Events records are official and hash-verified, but the public detail files do not provide machine-readable publication timestamps. The SACB therefore uses event start as an availability proxy, making event-conditioned results retrospective sensitivity analyses rather than operational forecasts.

The second limitation is geographic validation. The experiment covers one county-level PM2.5 aggregate from 2019 through 2025. The monitor-level analysis shows that event-retrieval effects change sign across sites, but it does not establish transfer to another county, climate, or emissions regime.

The third limitation is the composition of the holdout. Only 33 target-event origins and 37 threshold exceedances are observed, and no test origin exceeds 55.4 μg/m^3. Among the broader 70-origin event-context subset, the qualifying category analysis contains flood, heavy-rain, and other-weather contexts but no wildfire, smoke, or high-wind query. The January 2025 high-error period is validation evidence, not external confirmation. Claims about severe smoke episodes or operational alerts would therefore exceed the observed test support.

The fourth limitation is target construction. A county median reduces monitor missingness but smooths local peaks and depends on the active monitor roster. The lower-coverage monitor arms reveal heterogeneity but are not sufficiently dense for event-subset inference.

The fifth limitation concerns empirical strength. TRACE-RAF does not outperform Ridge, and neither its residual-gate contrast nor its event-channel contrast is significant after Holm adjustment. Evidence records are auditable, but the faithfulness perturbations for event and retrieval families are mixed and do not support causal interpretation.

The sixth limitation concerns model integration. LSER is a transparent similarity retriever rather than a learned dual encoder, and Chronos-Bolt is a frozen benchmark rather than an internal prompting backbone. The method should not be represented as a reproduction of TimeRAF's Channel Prompting mechanism.

Finally, hardware identifiers, peak memory, and a Git commit were unavailable inside the extracted run environment. Package versions, configuration hashes, archive hashes, and prediction-level verification improve reproducibility, but the reported runtime is not a hardware-normalized efficiency measurement.

X. FUTURE WORK

Operational event studies should replace event-start proxies with archives that expose genuine issue or publication times and should include smoke-specific sources when wildfire transport is the target mechanism. External validation should apply the frozen protocol to a second US county with distinct meteorology, emissions, and monitor density. Spatial or site-aware forecasting could then test whether county aggregation masks event effects. Longer-term work may evaluate a learned residual retriever, richer event embeddings, internally conditioned TSFMs, and calibrated exceedance probabilities. Each extension requires a pre-registered external holdout and component-wise ablation under the same temporal embargo.

XI. CONCLUSION

This study presented TRACE-RAF, a Trust-Gated Residual Analog Correction method for auditable 24-hour PM2.5 forecasting. SACB aligns official PM2.5, GHCNh weather, calendar, and Storm Events records; LSER enforces a strict historical embargo; TRACE retrieves out-of-fold residual analogues and applies a learned reliability gate; and DFEH retains prediction-level evidence and drift diagnostics. On the final 2025 holdout of 2,460 origins, TRACE-RAF obtains MSE 40.062, MAE 4.134, RMSE 6.329, and $R^2 = 0.444$. This improves context XGBoost and is nominally better than the LightGBM comparator, while Ridge remains strongest at MSE 39.144. The mechanism ladder shows that validation rejects both raw-future fusion and a global residual scale, while the learned gate conservatively reduces the residual update and provides a nominal MAE improvement. No favorable TRACE comparison survives Holm adjustment, and the event-free TRACE ablation is effectively unchanged. The supported contribution is therefore a reproducible, trust-gated residual-retrieval architecture and an auditable evaluation of event-conditioned residual retrieval and one frozen Chronos-Bolt-small output-fusion configuration; it is not a claim about retrieval-augmented TSFMs generally or universal baseline superiority. Strict event issue times, external geographic testing, and smoke-specific validation remain necessary before operational deployment.

REFERENCES

[1] World Health Organization, "WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide," 2021. [Online]. Available: https://www.who.int/publications/i/item/9789240034228

[2] A. Das, W. Kong, R. Sen, and Y. Zhou, "A decoder-only foundation model for time-series forecasting," in *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[3] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo, "Unified training of universal time series forecasting transformers," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 53 140–53 164.

[4] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski, "MOMENT: A family of open time-series foundation models," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 16 115–16 152.

[5] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," *Transactions on Machine Learning Research*, 2024.

[6] V. Ekambaram, A. Jati, P. Dayama, S. Mukherjee, N. H. Nguyen, W. M. Gifford, C. Reddy, and J. Kalagnanam, "Tiny time mixers (TTMs): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[7] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W. tau Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[8] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020, pp. 6769–6781.

[9] H. Zhang, C. Xu, Y.-F. Zhang, Z. Zhang, L. Wang, and J. Bian, "TimeRAF: Retrieval-augmented foundation model for zero-shot time series forecasting," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 9, pp. 5654–5665, 2025.

[10] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[11] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[12] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[13] K. Cho, B. van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 2014, pp. 1724–1734.

[14] H. Hewamalage, C. Bergmeir, and K. Bandara, "Recurrent neural networks for time series forecasting: Current status and future directions," *International Journal of Forecasting*, vol. 37, no. 1, pp. 388–427, 2021.

[15] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski, "DeepAR: Probabilistic forecasting with autoregressive recurrent networks," *International Journal of Forecasting*, vol. 36, no. 3, pp. 1181–1191, 2020.

[16] R. Godahewa, C. Bergmeir, G. I. Webb, R. J. Hyndman, and P. Montero-Manso, "Monash time series forecasting archive," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

[17] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 12, 2021, pp. 11 106–11 115.

[18] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Advances in Neural Information Processing Systems*, vol. 34, 2021.

[19] T. Zhou, Z. Ma, Q. Wen, X. Wang, L. Sun, and R. Jin, "FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 27 268–27 286.

[20] Y. Zhang and J. Yan, "Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting," in *Proceedings of the 11th International Conference on Learning Representations*, 2023.

[21] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, and M. Long, "iTransformer: Inverted transformers are effective for time series forecasting," in *Proceedings of the 12th International Conference on Learning Representations*, 2024.

[22] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "TimesNet: Temporal 2d-variation modeling for general time series analysis," in *Proceedings of the 11th International Conference on Learning Representations*, 2023.

[23] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 9, 2023, pp. 11 121–11 128.

[24] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *Proceedings of the 11th International Conference on Learning Representations*, 2023.

[25] S. Wang, H. Wu, X. Shi, T. Hu, H. Luo, L. Ma, J. Y. Zhang, and J. Zhou, "TimeMixer: Decomposable multiscale mixing for time series forecasting," in *Proceedings of the 12th International Conference on Learning Representations*, 2024.

[26] B. Lim and S. Zohren, "Time-series forecasting with deep learning: A survey," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 379, no. 2194, p. 20200209, 2021.

[27] K. Benidis, S. S. Rangapuram, V. Flunkert, Y. Wang, D. Maddix, C. Turkmen, J. Gasthaus, M. Bohlke-Schneider, D. Salinas, L. Stella, F.-X. Aubet, L. Callot, and T. Januschowski, "Deep learning for time series forecasting: Tutorial and literature survey," *ACM Computing Surveys*, vol. 55, no. 6, pp. 1–36, 2023.

[28] Y. Liu, H. Zhang, C. Li, X. Huang, J. Wang, and M. Long, "Timer: Generative pre-trained transformers are large time series models," in *Pro-*

ceedings of the 41st International Conference on Machine Learning, 2024, pp. 32 369–32 399.

[29] Y. Liu, G. Qin, X. Huang, J. Wang, and M. Long, "Timer-XL: Long-context transformers for unified time series forecasting," in *Proceedings of the 13th International Conference on Learning Representations*, 2025.

[30] X. Liu, J. Hu, Y. Li, S. Diao, Y. Liang, B. Hooi, and R. Zimmermann, "UniTime: A language-empowered unified model for cross-domain time series forecasting," in *Proceedings of the ACM Web Conference*, 2024, pp. 4095–4106.

[31] M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Li, S. Pan, and Q. Wen, "Time-LLM: Time series forecasting by reprogramming large language models," in *Proceedings of the 12th International Conference on Learning Representations*, 2024.

[32] T. Zhou, P. Niu, X. Wang, L. Sun, and R. Jin, "One fits all: Power general time series analysis by pretrained LM," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[33] J. Liu, L. Yang, H. Li, and S. Hong, "Retrieval-augmented diffusion models for time series forecasting," in *Advances in Neural Information Processing Systems*, 2024, pp. 2766–2786.

[34] Y. Ang, Y. Bao, Q. Huang, A. K. H. Tung, and Z. Huang, "TSGAssist: An interactive assistant harnessing LLMs and RAG for time series generation recommendations and benchmarking," *Proceedings of the VLDB Endowment*, vol. 17, no. 12, pp. 4309–4312, 2024.

[35] S. Zhou, W. Wang, L. Zhu, Q. Qiao, and Y. Kang, "Deep-learning architecture for PM2.5 concentration prediction: A review," *Environmental Science and Ecotechnology*, vol. 21, p. 100400, 2024.

[36] Z. Zhang and S. Zhang, "Modeling air quality PM2.5 forecasting using deep sparse attention-based transformer networks," *International Journal of Environmental Science and Technology*, vol. 20, pp. 13 535–13 550, 2023.

[37] Y. Liang, Y. Xia, S. Ke, Y. Wang, Q. Wen, J. Zhang, Y. Zheng, and R. Zimmermann, "AirFormer: Predicting nationwide air quality in china with transformers," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 12, 2023, pp. 14 329–14 337.

[38] US Environmental Protection Agency, "AirData: Pre-generated air quality data files," https://aqs.epa.gov/aqsweb/airdata/download_files.html, 2026, accessed: 2026-07-17.

[39] NOAA National Centers for Environmental Information, "Global historical climatology network-hourly (GHCNh), version 1," https://doi.org/10.25921/jp3d-3v19, 2023, accessed: 2026-08-23.

[40] ——, "Storm events database: Bulk data download," https://www.ncei.noaa.gov/stormevents/ftp.jsp, 2026, accessed: 2026-07-17.

[41] NOAA Office of Satellite and Product Operations, "Hazard mapping system fire and smoke product," https://ospo.noaa.gov/products/land/hms.html, 2026, accessed: 2026-07-17.

[42] Z. Liu, M. Cheng, Z. Li, Z. Huang, Q. Liu, Y. Xie, and E. Chen, "Adaptive normalization for non-stationary time series forecasting: A temporal slice perspective," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[43] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[44] C. Raman, A. Nonnemaker, A. Villegas-Morcillo, H. Hung, and M. Loog, "Why did this model forecast this future? information-theoretic saliency for counterfactual explanations of probabilistic regression models," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

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